TOPIC ..... DATE .....

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DBMS Assignment (Section - A, B.Tech CSE Core)

1. What is Database Management System (DBMS)? Why is it required?

=> A database management System (DBMS) is Software used to store, manage, retrieve and organize data efficiently.
It is required because:
- It ensures data security.
- Maintains data consistency.
- Reduce data redundancy.
- Provides backup and recovery.
- Supports multi-user access.
Eg; Banking system, hospital database.

2. Why is DBMS better than the traditional file System?

=> DBMS is better because:-
- Reduces Redundancy.
- Avoids data inconsistency.
- Provides security.
- Supports concurrency.
- offers backup & recovery.
- Easy data sharing.

3. What is Data Abstraction? Explain levels.

=> Data Abstraction means hiding internal details & sharing only necessary information to users.
Levels:-
Physical Level = How data is stored.
Logical Level = Structure of data.

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TOPIC ..... DATE .....

View Level = Users specific data.

4. What is three-schema architecture? Why is it needed?
=> it includes - internal level.
- Conceptual level.
- External level.

Importance
- Provides data independance
- Simplifies database design.
- Improves security.

5. What is Data Independance? Why is it needed?
=> It is the ability to change schema at one level without affecting others.
Types - Logical.
- Physical
-> Needed for flexibility & System maintenance.

6. What is Entity - Relationship (ER) models?
=> The Entity-relationship model represents real-world objects, attributes and relationships graphically.

7. What are integrity constraints? Why are they important in DBMS?
=> Rules that maintain accuracy and consistency:
- Primary Key.
- Foreign Key.
- Unique.
- Not Null.
- Check.

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TOPIC ..... DATE .....

Section - B :-

1. Draw the three-schema architecture and explain each level?

=>

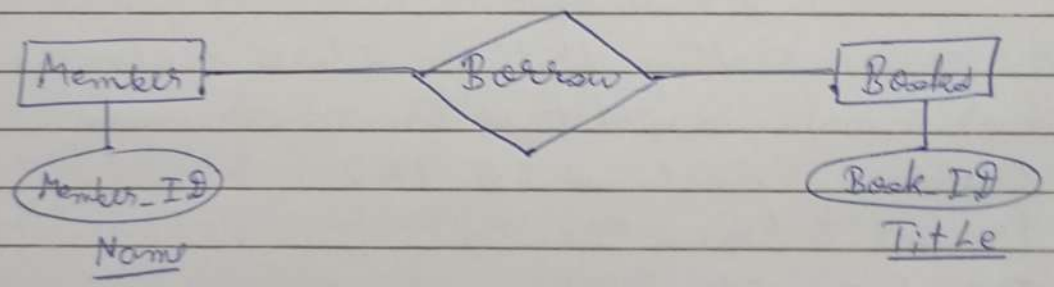
Users
↓
External Level (views)
↓
Conceptual Level (Logical Structure)
↓
Internal Level (Physical Structure)

External : User view
Conceptual : Overall database.
Internal : Storage details.

Benefit :- Data independence.

2. Draw an ER diagram for a Library Management System?

=>



Entities :- Member, Book.
Relationship :- Borrow.
Key :- Member_ID, Book_ID [one member can borrow many books].

3. Draw an ER diagram for Student-course registration system?

=> Student (Student ID, Name)
Course (Course ID, Course Name)
Relationship :- Student register for course.

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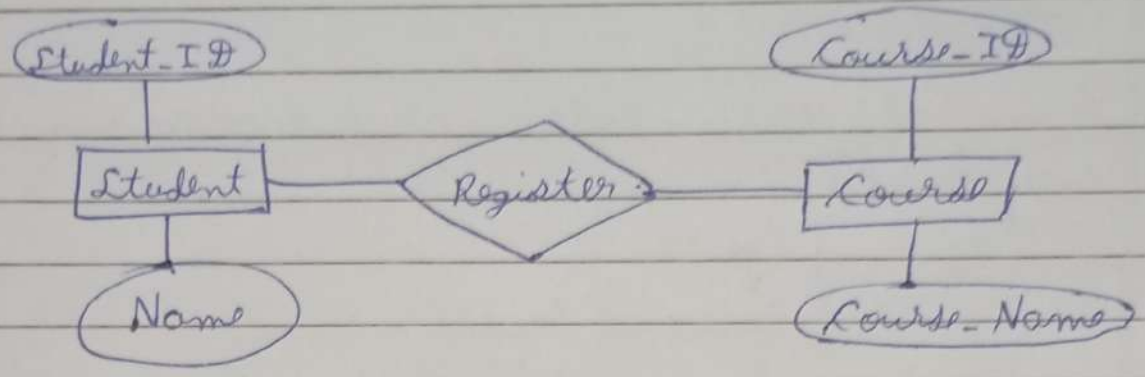
TOPIC ..... DATE .....



Section - C :-

1. A hospital want to store patient, doctor & appointment data. Identify entities, attributes and relationship?

=> Entities :- Patient, Doctor, Appointment, Patient books Appointment, Doctor attends Appointment.

Primary Keys :- Patient_ID, Doctor_ID, Appointment_ID

2. An online shopping platform stores customers, product & order data. Suggest keys & constraints?

=> Customer (Customer ID - PK).
Product (Product ID - PK).
Order (order ID - PK).
Foreign key in order table connect customer & product.

3. A University stores student records. Explain how abstraction help diff users?

=> Student see marks (view Level).
Admin manger records (Logical Level).
DBA controls storage (Physical Level).

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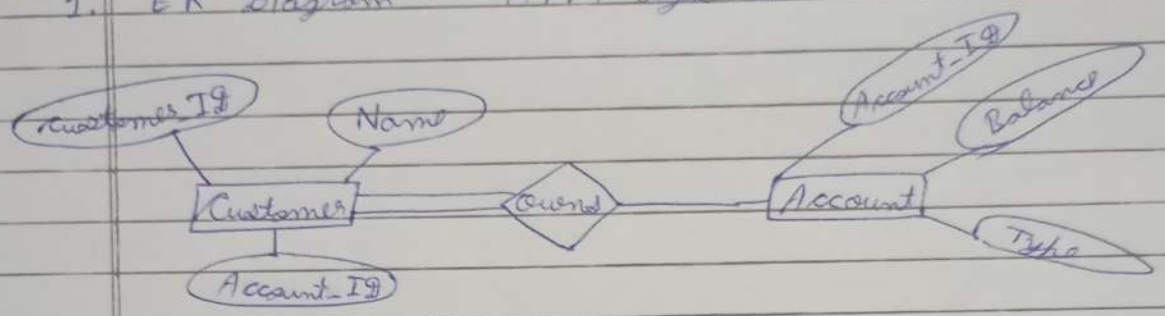
TOPIC ..... DATE .....

Section D :-

1. ER Diagram :- ATM System



Account-no. is primary key.
Relation : Customer owns Accounts.
Account no. take uniquely, Identifiers Account.

2. Three-Schema Architecture + one Benefit.
=> Benefit :- Provide Data Independence
change in Storage do not affect users.

External Level (user view)
↓
Conceptual Level (logical structure of Database)
↓
Internal Level (Physical Storage Level)

3. Draw a Entity with Attributes.
=> Entity! Employee
Employee-ID (PK)
Name
Salary
Department
Primary Key : Employee-ID.
Because of all employee their ID will be different.

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TOPIC ..... DATE .....

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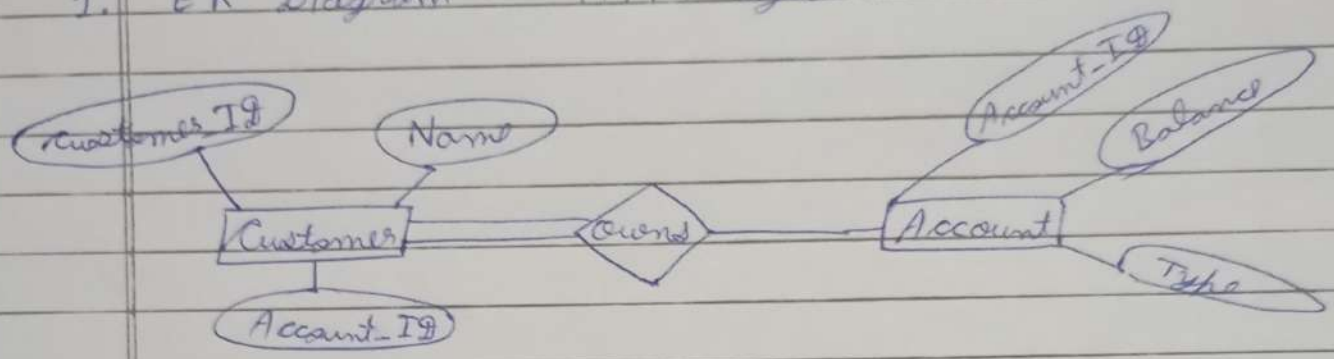

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